LITERATURE REVIEW

RETINAL DISEASE DETECTION USING DEEP LEARNING

MCM (Data Analytics)
Avinash Vadlamudi, Lohitha Vatti
A00053218, A00049933

1.INTRODUCTION

The causes of vision loss in the global community are retinal diseases including diabetic retinopathy (DR), age-related macular degeneration (AMD), glaucoma, and cataracts. It has been reported by WHO that approximately 2.2 billion individuals have some sort of visual impairment and most of such cases are preventable if they are diagnosed in time and treated accordingly [1], [8]. AMD is one of the major causes of blindness in older people, and it is reported to impact around 196 million individuals worldwide [4], [14] alone. These statistics underscore this increasing necessity of proper and prompt diagnosis of retinal diseases.

Conventional retinal diseases have been evaluated by manual reading of fundus images and Optical Coherence Tomography (OCT) scans by the ophthalmologists. It is a long and time-consuming process, despite being effective and it requires the experience of specialists. In most areas, there is a lack of access to trained professionals which pretends early diagnosis and treatment [1], [6]. Moreover, clinical decisions can be inconsistent with the manual assessment. Thus, the clinicians require automated and credible diagnostic systems to help them work less.

Deep learning methods have demonstrated a great potential in detecting retinal diseases in the past few years. Convolutional Neural Networks (CNNs), VGG, DenseNet, MobileNet and EfficientNet are all highly accurate in diagnosing and classifying DR and other retinal diseases [3], [6], [13]. These models are self-learned to identify significant retinal visual features such as lesions and abnormal patterns. Much more recently, models that use transformers like Vision Transformers (ViTs) and Swin Transformers have been developed in order to enhance performance by learning to learn relationships in the entire image [1], [2], [4], [11]. Multi-disease detection has also been suggested to be done using hybrid models, which are deep learning models that are trained alongside conventional machine learning models [7].

Although most deep learning systems are highly precise, most of them are black-box models, meaning that their decision-making mechanism cannot be comprehended easily [2], [5], [9]. This non-transparency presents clinical adoption difficulties as physicians must know the reason that a given system is making a specific prediction. To overcome this problem, Explainable Artificial Intelligence (XAI) models (Grad-CAM, SHAP, LIME and attention mechanisms) have been incorporated into retinal diagnostic methods [2], [3], [4], [5], [8], [9], [12], [13]. These techniques offer visual descriptions that indicate significant areas in retinal images. There is however some concern on the reliability and clinical validation of such explanations [9], [10].

This literature review examines current studies on automated retinal disease detection with reference to deep learning models, transformer-based models, explainable AI methods, and clinically structured grading systems. It will focus on examining the latest developments, trying to determine the existing limitations, and outline gaps in the research that should be bridged to create effective and valid retinal AI systems. This review is guided by the research question: What systematic approaches can be used to incorporate and clinically validate Explainable Artificial Intelligence (XAI) techniques in multiclass retinal disease classification systems to enhance trust, transparency, and clinical decision-making ?

2.BACKGROUND

The process of artificial intelligence in detecting retinal disease involves knowledge on medical imaging methods, deep learning algorithms, and explainable AI models. This part presents the major concepts and methods which are the basis of the existing studies in the area of automated retinal diagnosis.

2.1 RETINAL IMAGING MODALITIES

The two primary methods of imaging in retinal diagnosis are fundus photography and Optical Coherence Tomography (OCT). The fundus imaging is used to visualize the retina surface and to detect the presence of abnormalities like microaneurysms, hemorrhages and exudates in diabetic retinopathy (DR) [3], [6], [13]. The reason why fundus imaging is a popular screening tool is that it is a non-invasive, cost-effective screening tool.

OCT imaging can show cross-sectional images of retinal layers and I can observe structural alterations, fluid retention, and tissue damage. OCT has been found especially useful in the diagnostic/grading of age-related macular degeneration (AMD) [12], [14]. Automated grading systems have been implemented using large annotated OCT datasets in order to recreate clinical reasoning [14].

2.2 DEEP LEARNING FOR RETINAL DISEASE CLASSIFICATION

Convolutional Neural Network (CNN) or deep learning has become the prevalent model of retinal image analysis. VGG, DenseNet, MobileNet, EfficientNet, and ResNet are CNN models that learn hierarchical features on the images automatically [3], [6], [13]. The basic patterns that are detected in the early layers include edges and textures whereas the deeper layers recognize the complex disease-related structures.

Transfer learning in medical imaging not only to address limited dataset sizes but also to mitigate domain shift and improve model generalization. Retina images are finetuned on pre-trained models, which are trained on large depiction datasets to enhance their performance and decrease on training time [6], [7]. Research indicates that multiclass retinal disease detection using transfer learning is more accurate in classification [6].

Later, models based on transformers like Vision Transformers (ViTs) and Swin Transformers have been proposed [1], [2], [4], [11]. Transformers in contrast to CNNs incorporate self-attention mechanisms to learn long-range dependencies in the entire image. RetXplainer is a pretrained ViT that uses reinforcement learning to select the most informative patches of the image to classify the image into multiple labels as a retinal disease [1]. Fusion models that are based on attention have also been suggested to enhance robustness and feature representation [8], [11]. Hybrid methods incorporate CNNs feature extraction with conventional machine learning classifiers like the Support Vector Machines (SVM) and the Random Forest [7]. The objectives of these methods are to enhance stability of classification and deal with complicated disease patterns.

2.3 EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

Lack of transparency is one of the primary problems of medical diagnosis through deep learning. Most of the high-performing models are black boxes so that clinicians find it hard to place their confidence in them [2], [5], [9]. The interpretability is enhanced with the help of Explainable Artificial Intelligence

(XAI) methods. Explainable Artificial Intelligence (XAI) provides insight into decision-making process of deep learning models by highlighting the image regions that drive predictions. This is particularly important in medical imaging, where clinical trust and validation are important. Grad-CAM and Grad-CAM incremental methods produce heatmaps, which indicate regions of an image that have an impact on model prediction [2], [3], [8], [9], [12], [13]. SHAP and LIME are used to give both local and global explanations, by estimating the feature importance [4], [5], [8]. The visual insight of important regions of retinal images is also provided through attention mechanisms [10], [11].

Nevertheless, when a review of CAM-based methods of explanation in DR is conducted, it can be established that a large number of visual explanations can seem meaningful and not necessarily confirmed by clinical findings [9]. Other heatmaps are based on the blood vessels or bright areas rather than actual lesions. Thus, explainability is a significant area of research that should be evaluated.

2.4 MEASUREMENT OF RESULTS AND DATA

The models of retinal diseases are assessed in terms of accuracy, precision, recall, F1-score, ROC-AUC, sensitivity and specificity [2], [6], [7], [14]. In segmentation tasks of OCT, the overlaps between the predicted and true regions are measured using Dice Similarity Coefficient [14]. Popular datasets are EyePACS, Messidor, APTOS, IDRiD, RFMiD, and CATT [2], [4], [7], [9]. Numerous studies are based on publicly available datasets, yet the amount and the variety of data have been the weakness of a number of studies [1], [4], [11]. Overall, deep learning, transformer architectures, hybrid models, and XAI techniques form the core technical foundation of modern retinal disease detection systems.

3.LITERATURE REVIEW

The current studies in retinal disease detection can be grouped into four key themes, including (1) CNN and transfer learning models, (2) transformers and attention models, (3) explainable AI integration, and (4) multi-disease and clinically structured systems. In this section, there is the critical review of these approaches and comparison of their strengths and weaknesses.

3.1 CNN AND TRANSFER LEARNING APPROACHES

Convolutional Neural Networks (CNNs) have remained popular models in classification of retinal images. A number of research papers show that CNN models, including MobileNet, DenseNet, VGG, and EfficientNet can be effectively used to diagnose diabetic retinopathy (DR) and other retinal pathologies.

Deep learning model with MobileNet and DenseNet demonstrated high accuracy in detecting severity with DenseNet performing 93 percent in classifying single class which is diabetic retinopathy and severity levels using fundus images [3]. The paper has drawn attention to the fact that CNNs are useful in detecting the patterns of lesions but pointed out the necessity of explainability to achieve clinical trust. As research progressed, studies began addressing multiclass classification, which are realistic towards the real-world scenarios. A comparative analysis of EfficientNet, DenseNet121, and VGG16 evaluated multiclass fundus images classification across Normal, Diabetic Retinopathy, Glaucoma, Cataract categories [6]. In this setting, EfficientNet obtained an accuracy of 92.26%. Unlike binary classification tasks, multiclass

frameworks must distinguish between multiple disease categories simultaneously, increasing model complexity and diagnostic difficulty.

Although the accuracy reported in [3] 93 percent appears slightly higher than that in [6] 92.26 percent, the tasks differ significantly. That paper focused on disease severity grading within specific pathology, whereas the latter addressed broader multiclass disease categorization. Therefore, direct numerical comparison should be interpreted cautiously due to differences in dataset composition, class distribution and classification objectives.

In another research, A transparent diabetic retinopathy (DR) detection model based on a CNN architecture with LIME explanations achieved 94% accuracy in a binary classification setting [5]. Although the reported performance was strong, the relatively small dataset limited its generalization capability. To improve robustness, a hybrid framework combining VGG16 feature extraction with Support Vector Machine (SVM) and Random Forest classifiers was evaluated on the IDRiD dataset, achieving an F1-score of 0.93 and ROC-AUC of 93.12% [7]. This hybrid approach demonstrated improved classification stability compared to single CNN-based models. Nevertheless, while CNN-based systems achieve high accuracy, they primarily focus on local feature extraction and may struggle to capture long-range dependencies in complex retinal images. Furthermore, many CNN studies emphasize performance metrics over interpretability, which limits clinical transparency and practical trust in real-world settings [6], [7].

3.2 TRANSFORMER AND ATTENTION-BASED ARCHITECTURES

Transformer-based architectures have been proposed in order to overcome CNNs limitations. ViTs and Swin Transformers apply the notion of self-attention to images to extract global contextual relationships in images.

RetXplainer introduced an explainable model comprising of a pretrained Vision Transformer and reinforcement learning to classify retinal diseases that have multiple labels [1]. The model gave ROC-AUC 0.9110 and adopted patch selection approach to indicate diagnostically significant areas. This method enhanced the interpretability by directly choosing significant image patches. Nevertheless, the research recognized the limitations of the dataset size that can have an impact on generalization.

A CNN-transformer architecture with CNN and transformer models were used to detect the DR and obtained AUC-ROC of 0.96 on the benchmark data like EyePACS and Messidor [2]. The model also facilitated use on edge equipment. The performance was also good though explainability was based primarily on the Grad-CAM and SHAP visualization. The CNN, MLP, and transformer classifiers were tested in a cascade-based AMD detection framework [4]. The CNN recorded the best accuracy (94.19%), and the transformer classifier was not as accurate in this particular case. It indicates that transformers might need bigger datasets or specialized optimization methods in order to achieve better results than CNNs in medical imaging.

MedFusion-XAI proposed a refined attention and gated fusion Swin Transformer design of medical image classification [11]. The research was designed to enhance interpretability and consistency of attention. Nevertheless, the validation was reduced to particular datasets and cross-data robustness was not

thoroughly assessed. On the whole, transformer-based models have better global feature representation compared to CNNs, though they usually need large datasets and optimization. Their superiority when compared to CNNs in small medical data is still unclear. Building on these developments, a comprehensive analysis of CAM-based explanation methods in diabetic retinopathy emphasizes the growing role of visual explanation techniques in enhancing model transparency and clinical interpretability [15].

3.3 EXPLAINABLE AI IN RETINAL IMAGING

Explainability is a significant research concern in AI systems in the retina. Several high-performing models are used as black boxes and introduce problems with clinical adoption [2], [5], [9]. The most popular method of visual explanation is Grad-CAM. Experiments that combine Grad-CAM with CNN models demonstrate better transparency with respect to highlighting important areas of the retina [2], [3] and [13]. EDDNet30 used spatial attention and multi-scale fusion modules with Grad-cam and LIME with an accuracy of 95.29 on nine types of eye diseases [8]. In this research, there was high performance in multiclass and consideration of interpretability.

Nevertheless, when examining the CAM-based approaches to DR, the serious flaws of the existing explainability techniques were pointed out [9]. Most studies base their studies on visual heat maps only, with no validation on whether highlighted areas reflect actual lesions or not. The review has highlighted the fact that lesion-level ground truth or clinical validation is rarely used to assess the quality of explanation.

A dual-branch attention network that includes Grad-CAM interpretability demonstrated better classification behavior but viewed explainability primarily as an added post-hoc feature, and not as a contentious training goal [10]. Equally, OCT-based explainable classification models showed alignment with clinical features visually but did not have systematic quantitative assessment of interpretability [12]. Despite the fact that XAI enhances transparency, the majority of the approaches are still visualization-based and do not deal with the issues of causal faithfulness or strongness in full. Thus, the field of explainability research has yet to gain better assessment tools and clinical participation.

3.4 MULTI-DISEASE AND CLINICALLY STRUCTURED SYSTEMS

In addition to detecting individual diseases, there are also studies which examine multi-disease or clinically summarized diagnostic systems. A system of multi-disease detection with the retinal, pulmonary, cardiovascular, and renal data that unites both deep learning and machine learning models was used [7]. VGG16 using hybrid classifiers had good F1-score and ROC-AUC when used to classify retinas. Nevertheless, it is still difficult to combine heterogeneous sources of data because of the problem of feature alignment.

Multi-class eye disease detection using better dataset diversity and augmentation methods were also covered by EDDNet30 [8]. It showed good performance yet it still depended on visual explainability techniques. An OCT B-scan-based system of grading AMD clinically explainable was suggested and evaluated in a structured way, imitating physician reasoning [14]. The system segmented and detected layers prior to final classification and gave clinically meaningful reports on why grading decisions were

made. This system combined interpretability with the diagnostic pipeline, as opposed to the heatmap-based elucidations. Nevertheless, implementation on actual hospital systems is a viable issue.

These clinically organized systems are the transformation of the performance based models to decision-support tools that are more compatible with the medical workflows. Quite however, the main issue is scalability, variety of datasets, and their integration into healthcare systems.

3.5 Critical Comparison of the Performance

In the literature, both CNN and transfer learning methods have shown good performance and efficiency albeit with no global context modelling. Transformer based models have superior attention mechanisms, but typically require larger datasets and fine tuning. Hybrid systems enhance robustness at the expense of complexity in architecture. Explainable AI methods have been used extensively but the majority of the explanations are visual and have not been scientifically tested in clinical settings. Multi-disease and clinically structured systems have potential in matching AI models to the real world but need more testing across the different populations. The future studies ought to be dedicated to uniting powerful transformer designs, standard explainability assessment, multi-institutional dataset testing, and clinically valuable output design.

4.GAP ANALYSIS AND RESEARCH OPPORTUNITIES

It is true that much progress has already been achieved in the field of automated retinal disease recognition, but there are still some important gaps in the existing literature. These gaps are associated with dataset constraints, model generalizability, explainability validation, and the integration of the model in the clinic.

Data size and heterogeneity is one of the greatest limitations in most studies. Some of the works used relatively small or single source datasets, which might tend to limit the generalization to larger populations [1], [4], [11]. Although models are, in many cases, assessed to be within the same dataset distribution when using public datasets like EyePACS, Messidor, APTOS, IDRiD, or CATT [2], [4], [7], [9]. This presents the issue of reliability in the performance of various imaging equipment, population groups, and clinical settings. Reported high accuracy can be irrelevant in the real world without multi-center validation.

The other significant issue with the dataset of retinal disease is class imbalance. In some publicly available datasets, such as EyePACS and APTOS, the count of normal or mild cases is much greater than the count of severe diseases. This bias may cause models to give an overall bias towards the majority classes, and overstatement of the overall accuracy at the cost of sensitivity to clinically significant, but infrequent conditions. Such techniques are necessary in a balanced performance such as weighted loss, focal loss, oversampling and data augmentation.

One of the other gaps is associated with explainability assessment. Although some studies combine Grad-CAM, SHAP, LIME, or attention mechanisms to give visual explanations [2], [3], [5], [8], [13], explanation quality is not frequently calculated with the help of quantitative or clinical validation. An overview of CAM-based visual explanations also indicated that heatmaps tend to pay attention to the non-diagnostic areas like blood vessels or bright objects rather than actual lesions [9]. Arrange of explanations in most instances is like a visual assurance, but not a strictly proven evidence of model thinking [10], [12].

No universal metric of measuring explanation faithfulness, stability or clinical usefulness currently exists. Such an absence of appraisal curtails trust and feasible implementation.

Models that use transformers also pose other research problems. Even though Vision Transformers and Swin Transformers are excellent at capturing global dependencies [1], [2], [11], a number of studies have demonstrated that CNN-based models remain superior to transformers in some medical contexts, in particular, where data is slight [4]. In clinical research, large datasets and computational power may be required to be used with transformers, and this may not be accessible. More effort is required in creating light transformer architectures and data-efficient transformer architecture that can be implemented in medical imaging.

Multi-disease and hybrid systems are also complex. Hybrid methods that use deep learning-based ones with conventional machine learning classifiers have shown promising results [7], yet mixing heterogeneous types of data is complicated. The problem of feature alignment of images and structured clinical data remains an open problem. Additionally, with more complex models, interpretability can decline, and more often it will be necessary to rely on post-hoc methods of explanation.

Another gap is that of clinical integration. Certain works suggest the use of edge-AI and cloud-based architecture [2], [5], yet practical application needs to be adjusted to the needs of hospital information systems, regulatory regulations, and workflow restructuring. Even systems structured in a clinical way, like the OCT-based AMD grading models simulating physician reasoning [14] still need additional validation across different hospitals. Practical problems like variation in image quality, uncommon diseases and real-time performance limitations should be resolved before it gains widespread use.

On the basis of these gaps, there are a number of research opportunities. To begin with, future research ought to aim at putting a greater emphasis on multi-center and cross-dataset validation to achieve robustness and fairness. Second, metrics of standardized explainability evaluation need to be created such as lesion-level validation and clinician-based assessment protocols. Third, the studies need to be conducted on lightweight transformer architectures and data-efficient training strategies. Fourth, incorporating multimodal data, e.g. uniting fundus data, OCT scans, and clinical data, can enhance the accuracy of diagnosis and also represent the practice in the real world [7], [14]. Lastly, AI researchers and ophthalmologists should work more closely to make sure that the models generate clinically meaningful results instead of high performance measures. Sealing such gaps will enable building of verifiable, transparent and scalable retinal AI systems that can be practically used in the clinical setting.

5.CONCLUSION

The creation of deep learning, transfer learning and transformer-based architectures have made automated retinal disease detection much more advanced. DenseNet, MobileNet, EfficientNet, and VVG are convolutional neural networks that have shown great capability in the detection and classification of diabetic retinopathy and other retinal diseases [3], [6], [13]. Transformer models such as Vision Transformers and Swin Transformers have also enhanced representation of global features and attention modeling in retinal images [1], [2], [11]. Multi-disease classification tasks have also produced promising

results using hybrid systems which are a combination of deep learning and traditional machine learning methods [7].

Besides the performance enhancement, explainability has become a significant point of research. Grad-CAM, SHAP, LIME, and attention mechanisms are examples of the Explainable Artificial Intelligence (XAI) approaches that have been incorporated in numerous studies to offer visual explanations of model forecasts [2], [3], [4], [5], [8], [12]. Clinically organized systems, especially systems of OCT-based grading, strive to recreate the way physicians think and offer more insightful comments [14]. These achievements are significant milestones of enhancing transparency and gaining the trust of clinicians.
Nevertheless, there are still a number of limitations. Most models are trained and tested on small datasets or single-source datasets, which limits generalizability [1], [4], [11]. Explainability techniques have a tendency to undergo visual validation as opposed to the standard clinical evaluation [9], [10]. Transformer-based models can be intensive with respect to the size of datasets and the computational demands, which preclude their practical application in certain settings [4]. Moreover, implementation into the actual clinical workflows needs further validation, regulation and system compatibility.

In general, the existing studies prove that AI-based diagnosis of retinal diseases has a great potential, but additional research is needed to guarantee its strength, transparency, and clinical usability. The future studies shall focus on multi-center validation, standardized explainability assessment, light and efficient model design, and enhanced cooperation among AI researcher and ophthalmologists. Overcoming these challenges will allow retinal AI systems to transition through research prototypes to credible clinical decision-support systems to enhance early diagnosis and patient results.

6. REFERENCES

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